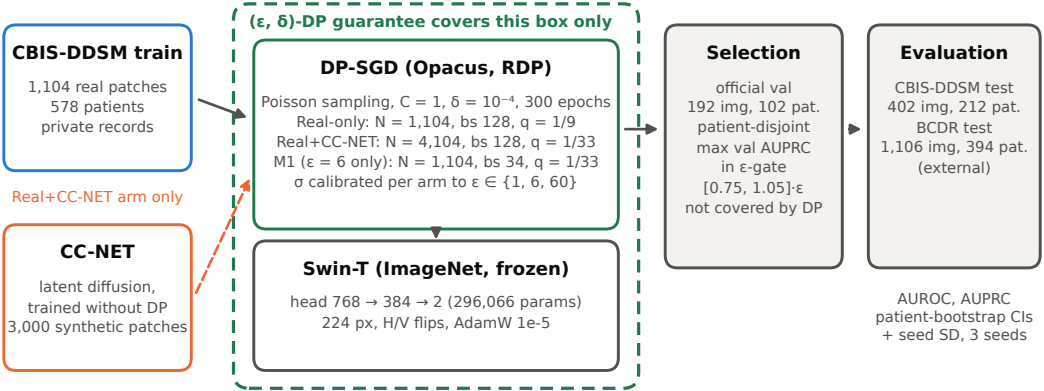
CBIS-DDSM train
1,104 real patches
578 patients
private records
Real+CC-NET arm only
CC-NET
latent diffusion,
trained without DP
3,000 synthetic patches
(ε, δ)-DP guarantee covers this box only
DP-SGD (Opacus, RDP)
Poisson sampling, C = 1, δ = 10⁻⁴, 300 epochs
Real-only: N = 1,104, bs 128, q = 1/9
Real+CC-NET: N = 4,104, bs 128, q = 1/33
M1 (ε = 6 only): N = 1,104, bs 34, q = 1/33
σ calibrated per arm to ε ∈ {1, 6, 60}
Swin-T (ImageNet, frozen)
head 768 → 384 → 2 (296,066 params)
224 px, H/V flips, AdamW 1e-5
Selection
official val
192 img, 102 pat.
patient-disjoint
max val AUPRC
in ε-gate
[0.75, 1.05]·ε
not covered by DP
Evaluation
CBIS-DDSM test
402 img, 212 pat.
BCDR test
1,106 img, 394 pat.
(external)
AUROC, AUPRC
patient-bootstrap CIs
+ seed SD, 3 seeds